CHROMATIC NULLSTELLENSATZ LECTURE 1

TYPED BY AOWEN FAN

In the first lecture given by Guchuan Li, we begin with some basic setups for chromatic homotopy theory. We then explain the main results of [BSY22], the idea behind them and their applications without presenting any proof.

1. Chromatic homotopy theory

In the chromatic world, we fix a prime p and a height $n \in \mathbb{N}$. The central object is the Lubin-Tate theory.

Theorem 1.1 ([GH04]; [Rez98])**.** *Given a pair $(k, \mathbb{G})$, where k is a perfect field of character p and $\mathbb{G}$ is a formal group of height n, there is an $\mathbb{E}_\infty$-ring $E_n(k, \mathbb{G}) \in \mathrm{CAlg}(\mathrm{Sp}_{K(n)})$, such that $E_n(k, \mathbb{G})$ is even periodic and $\pi_0 E_n(k, \mathbb{G})$ can be identified with the Lubin-Tate ring.*

Throughout this seminar, we focus on an algebraic closed field $k = \overline{k}$, in which case there is only one formal group up to isomorphism. Applying the above construction, we obtain $E_n(k)$ called **height n Lubin-Tate** theory. The homotopy group can be characterized as

$$\pi_*(E_n(k)) = W(k)[\![u_1, \cdots, u_n]\!][v^{\pm 1}]$$

where $W(k)$ is the Witt ring of k, $u_1, \cdots, u_n$ are in degree 0 and v is in degree -2.

Another class of spectra is also of great interest, which is denoted by $K(n)$ and is called **Morava K-theory**, with homotopy ring given by

$$\pi_*(K(n)) = \mathbb{F}_p[v_n^{\pm 1}] \text{ where } |v_n| = (2p^n - 2)$$

It admits a structure of "field of ring spectra". But it has merely an $\mathbb{E}_1$-structure. One can work with $K(n)$-local category $\mathrm{Sp}_{K(n)}$. Especially, one has

$$\mathbb{S}_{K(n)} \simeq E_n(\overline{\mathbb{F}}_p)^{hG_n}$$

where G_n denotes the Morava stablizer group.

2. $T(n)$-local category

We now define $\mathrm{Sp}_{T(n)}$, which is the Bousfield localization of the " finite type-n-complex".

Proposition 2.1. *Let X be a finite p-local spectrum. Then $K(n)_*X = 0$ implies $K(n-1)_*X = 0$.*

Proof. See for example [Lur25][§26; Lemma 4]. □

Say a spectrum X is of type n if $K(n)_*X = 0$ and $K(n-1)_*X \neq 0$. It exists(also see [Lur25][§26]) and the $T(n)$-local category $\mathrm{Sp}_{T(n)}$ is defined as the X-local category, which does not rely on the choice of X.

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3. The Nullstellensatz

The classical Hilbert Nullstellensatz says that suppose k is algebraic closed and $f_1, \cdots, f_n \in k[x_1, \cdots, x_m]$, if the equation system $f_i = 0$ has a solution, then there exists $g_1, \cdots, g_n$ such that $g_1 f_1 + \cdots + g_n f_n = 1$. This can generalize to abstract settings.

Definition 3.1. (Nullstellensatz) Let $\mathcal{C}$ be a presentable category. We say an object $k \in \mathcal{C}$ is **nullstellenstatz** if for any compact nonterminal object $A \in \mathcal{C}_{k/-}$, the map $k \to A$ admits a retract.

Back to $k[x_1, \cdots x_n]$, if we consider the polynomial function system $(f_1, \cdots, f_n)$ with a solution and let $A = k[x_1, \cdots x_m]/(f_1, \cdots, f_n)$. The existence of solution guarantees that we can find a retraction $k \to A \to k$.

The main results can be stated as follows.

Theorem 3.2 (Theorem A)**.** *Let $\mathcal{C} = \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then the nullstellensatz objects are of the form $E(L)$, the Lubin-Tate theory attached to an algebraic closed field L.*

Theorem 3.3 (Theorem D)**.** *Suppose R is a $T(n)$-local $\mathbb{E}_\infty$-ring, then there exists $R \to E(L)$ for some L.*

This immediately proves a theorem of Hahn.

Theorem 3.4 ([Hah16])**.** *$R \in \mathrm{CAlg}(\mathrm{Sp}_p)$, then $R \otimes T(n) = 0$ implies $R \otimes T(n+1) = 0$. Similarly for $K(n)$.*

Hahn's result makes it possible for us to define the height of $R \in \mathrm{CAlg}(\mathrm{Sp})$. We say that R is of height n if $R \otimes T(n) \neq 0$ but $R \otimes T(n+1) = 0$.

Theorem 3.5 (Redshift)**.** *Suppose $R \in \mathrm{CAlg}(\mathrm{Sp})$ is nonzero, then $\mathrm{ht}(K(R)) = \mathrm{ht}(R) + 1$.*

The proof can reduce to the case where $R = E(L)$, which reduces to a theorem of [Yua24] that states $\mathrm{ht}(K(E(L))) = \mathrm{ht}(E(L)) + 1$.

4. $\mathbb{E}_\infty$-complex orientation

Question: Does the complex orientation $\mathrm{MU} \to E_n(L)$ admit $E_n(L)$ an $\mathbb{E}_\infty$-complex orientation?

In order to solve this, we will need to construct the so called univeral orientability. The answer to the question in this paper is

Theorem 4.1 (Theorem G)**.** *There exists an $\mathbb{E}_\infty$-complex orientation $\mathrm{MU} \to E_n(\overline{\mathbb{F}}_p)$.*

Theorem D cannot complete the proof of the above theorem. Since it only makes sure the existence of some extension $R \to E_n(L)$ for some unpredicted L. Therefore, we will need the following variant.

Theorem 4.2 (Theorem F)**.** *Fix L to be an algebraic closed field and $f : X \to \mathrm{pic}(\mathbb{S})$. Then the followings are equivalent*

(1) There exists an $\mathbb{E}_\infty$-complex orientation $Mf \to E(L)$.
(2) $L_{K(n)} Mf \neq 0$.

In particular, if we take $X = \mathrm{BU}$ and apply Theorem F, then we will obtain Theorem G.

Remark 4.3. By Hopkins-Smith's nilpotence theorem [DHS88], MU detects nilpotence. That means, $f : R_1 \to R_2$ is nilpotent (in the sense of [HS98][Definition 1]) if and only if $f_* : \mathrm{MU}_* R_1 \to \mathrm{MU}_* R_2$ is nilpotent.

Theorem 4.4 (Strict picard group)**.** *Let L be an algebraic closed field, then we have an equivalence*

$$\underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathrm{Mod}_{E(L)}\mathrm{Sp}_{T(n)})) \simeq \Sigma^{n+2}\mathbb{Z}_p \oplus \Sigma L^\times$$

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LECTURE 2

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In this lecture, we recall the construction of Lurie's spherical Witt ring and the Lubin-Tate theory attached to a perfect ring for a fixed character p, a finite height n and a given formal group $\mathbb{G}$ of height n. We combine these data into an adjunction, which is refered to as Lubin-Tate-tilt adjunction in [BSY22]. We explore the property of this adjunction, setting the stage for the next talk.

Introduction

Our main goal in this talk is to establish the following adjunction.

Theorem 0.1. *Fix a perfect field k of character p and a height n, then there is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

rendering Perf_k *as a colocalization of* $\mathrm{CAlg}^{\wedge}_{E(k)}$. *Here* Perf_k *denotes the category of (ordinary) perfect k-algebras and* $\mathrm{CAlg}^{\wedge}_{E(k)} := \mathrm{CAlg}(\mathrm{Sp}_{T(n)})_{E(k)/-}$ *is the category of $T(n)$-local $E(k)$-algebras.*

The main inputs are twofold. First, we need Lurie's spherical Witt functor, which instinctively owns a right adjoint.

$$\mathbb{W}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathrm{Sp}_p) : (-)^{\flat}$$

For any p-complete presentable stable ∞-symmetric monoidal category $\mathcal{C} \in \mathrm{CMon}(\mathrm{Pr}^{\mathrm{L}}_{st})$, we have the unit functor $\iota : \mathrm{Sp} \to \mathcal{C}$ inducing the adjunction

$$\iota^* : \mathrm{CAlg}(\mathrm{Sp}_p) \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : \iota_*$$

After combining these two adjunctions and taking $\mathcal{C} = \mathrm{Mod}^{\wedge}_{E(k)}$, we will get an adjunction between $\mathbb{W}(-)$ and $(-)^{\flat}$.

$$\mathbb{W}(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

Secondly, Lubin-Tate theories allow us to functorially assign each pair $(A, n, \mathbb{G}_0)$ a $T(n)$-local $\mathbb{E}_\infty$-ring $E(A, n, \mathbb{G}_0)$. Once we fix a given pair $(k, n, \mathbb{G}_0)$ such that A takes values in perfect k-algebras, we obtain a map $E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}^{\wedge}_{E(k)}$. We then compare $E(-)$ with $\mathbb{W}(-)$.

1. Thickenings and Spherical Witt rings

Recall that in classical Witt ring theory, we want to lift a ring A of character p to some universal $W(A)$ such that p is not a zero-divisor and $W(A)/pW(A) \cong A$. The Witt ring can be characterized by some universal properties. One can also construct $W(A)$ in a concrete way. For example, the underlying set of $W(A)$ consists of the formal series

$$[a_0] + [a_1]p + [a_2]p^2 + \cdots$$

and the multiplication is given by Witt polynomials.

The spherical Witt functor wants to generalize this construction to $\mathbb{E}_\infty$-settings. For example, the standard equivalence $\mathbb{Z}_p/p \xrightarrow{\cong} \mathbb{F}_p$ identifies $\mathbb{Z}_p$ with $W(\mathbb{F}_p)$. In spherical cases, we want to replace $\mathbb{Z}_p$ by $\mathbb{S}_p$. In order to make this spirit more precise, we need to introduce the concept of thickenings.

1.1. **Thickenings.**

Definition 1.1. Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ a finitely generated ideal, $f_0 : \pi_0(A) = A_0 \to B_0$ a map of ordinary rings. We say that $f : A \to B$ is an A-thickening of f_0 if we have the following square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

together with the following data

(1) B is an I-complete $\mathbb{E}_\infty$-ring.
(2) The above square induces an equivalence $\pi_0(B)/I \cong B_0$.
(3) For any $\mathbb{E}_\infty$-A-algebra R, there is a natural equivalence $\mathrm{Map}_A(B, R) \simeq \mathrm{Hom}_{A_0}(B_0, \pi_0(R)/I)$.

Example 1.2. Let $A = \mathbb{Z}_p$ and B_0 be an $A_0 = \mathbb{F}_p$-algebra. Then the $\mathbb{Z}_p$-thickening of f_0 is identified with the classical Witt ring $W(A)$.

Remark 1.3. By the universal property, the thickening is unique up to equivalence when it exists.

Lurie's theorem tells us that the thickening exists under some perfect assumptions, which covers the most interesting cases.

Theorem 1.4 ([Lur17a], Theorem 5.2.5)**.** *Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ be a finitely generated ideal, $f_0 : A_0 \to B_0$ be a relative perfect map between perfect $\mathbb{F}_p$-algebras. Then*

(1) The A-thickening of f exists.
(2) The thickening square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

is a pushout square of $\mathbb{E}_\infty$-rings.

In fact, in Lurie's proof, B can be constructed canonically as a filtered colimit of maping spaces, see [Lur17a, Section 5.2.1].

Example 1.5. Let $A = \mathbb{S}_p$ and $I = (p) \in A_0 = \mathbb{F}_p$. Let B_0 be a perfect $\mathbb{F}_p$-algebra. By the theorem, the $\mathbb{S}_p$-thickening of B_0 exists and we denote it by $\mathbb{W}(B)$. One can check that the construction is functorial, hence determines a functor

$$\mathbb{W}(-) : \mathrm{Perf} \to \mathrm{CAlg}(\mathrm{Sp}_p)$$

which we refer to as **spherical Witt functor**.

Before we continue to form the right adjoint, we introduce a detection for thickenings.

Proposition 1.6. *Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ be a finitely generated ideal, $f_0 : A_0 \to B_0$ be a relative perfect map between perfect $\mathbb{F}_p$-algebras. Then $f : A \to B$ is identified with the A-thickening of f_0 if and only if the following conditions are satisfied*

(1) B is an I-complete $\mathbb{E}_\infty$-algebra
(2) $\pi_0(B)/I \cong B_0$
(3) The thickening square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

is a pushout square of $\mathbb{E}_\infty$-rings.

Proof. The "only if" is part of the theorem's statement, we prove the inverse. Suppose B satisfies the above data, we can identify $\pi_0(B)/I$ with B_0. By the theorem, suppose $f : A \to D$ is the A-thickening of f_0, by the universal property that $\mathrm{Map}_A(D, B) \simeq \mathrm{Hom}_{A_0}(B_0, \pi_0(B)/I)$, one can lift the identity map on B_0 to the following square

$$\begin{array}{ccccc} A & \xrightarrow{f} & D & \xrightarrow{g} & B \\ \downarrow & & \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 & \xrightarrow{\mathrm{id}} & B_0 \end{array}$$

The outer square is a pushout square by the third assumption and the left square is a pushout square by the theorem. Therefore, the right square is a pushout square and g is forced to be an equivalence. □

Corollary 1.7. *The essential image of $\mathbb{W}(-)$ consists of those $\mathbb{E}_\infty$-ring R such that $R \otimes_{\mathbb{S}_p} \mathbb{F}_p$ is discrete.*

We can now form the right adjoint of $\mathbb{W}(-)$.

Proposition 1.8. *We define the **tilt functor** $(-)^\flat : \mathrm{CAlg}(\mathrm{Sp}_p) \to \mathrm{Perf}$ as $(-)^\flat = (\pi_0(-)/p)^\flat$. Then $(-)^\flat$ serves as the right adjoint for $\mathbb{W}(-)$.*

Proof. By definition, one has $\mathbb{W}(A)^\flat \cong A$. We can choose the unit map as the natural equivalenceid $\xrightarrow{\simeq} \mathbb{W}(-)^\flat$. The unit map induces

$$\mathrm{Map}(\mathbb{W}(A), R) \simeq \mathrm{Map}(\pi_0\mathbb{W}(A)/p, \pi_0 R/p) \simeq \mathrm{Map}((\pi_0\mathbb{W}(A)/p)^\flat, (\pi_0 R/p)^\flat) \simeq \mathrm{Map}(A, R^\flat)$$

where the first equivalence is from definitions of $\mathbb{W}()$ and thickenings and the second is because $\pi_0\mathbb{W}(A)/p$ is perfect. □

Example 1.9 (examples of spherical Witt rings)**.** Let M be an ordinary monoid. Then the spherical Witt ring of the perfect $\mathbb{F}_p$-algebra $\mathbb{F}_p[M]$ is exactly $\mathbb{S}_p[M]$. In particular, if we take $M = \mathbb{N}[1/p]$, we have $\mathbb{W}(\mathbb{F}_p[t^{1/p^\infty}]) \simeq \mathbb{S}_p[t^{1/p^\infty}]$.

1.2. **Properties of the spherical Witt functor.** We now list some properties of the spherical Witt functor.

Lemma 1.10. *For any $\mathbb{E}_\infty$-algebra A, the spherical Witt ring $\mathbb{W}(A)$ is p-adic formally étale, i.e. the relative cotangent complex $L_{\mathbb{W}(A)/\mathbb{S}_p}$ vanishes.*

Proof. This follows from the base change formula of cotangent complex along the pushout square that defines $\mathbb{W}(A)$. □

We will not use too much about the cotangent complex. In fact, we are interested in the following consequence of formally étaleness.

Proposition 1.11 ([Lur17b]; Remark 7.4.1.8)**.** *Let R be a p-adic formally étale $\mathbb{E}_\infty$-ring, then any square-zero extension $A \to B$ induces an equivalence*

$$\mathrm{Map}(R, A) \simeq \mathrm{Map}(R, B)$$

Remark 1.12. One can check [Lur17b, Section 7] for more details about square-zero extensions and cotangent complex. We will not discuss much here. But we do concern the following two examples of square-zero extensions. First, let R be a connective $\mathbb{E}_\infty$-algebra, each connection map in the Postnikov tower of R

$$\cdots \to \tau_{\leqslant 2}R \to \tau_{\leqslant 1}R \to \tau_{\leqslant 0}R$$

is a square-zero extension. Secondly, for any I-adic ordinary ring A, each connection map in the$I$-adic tower of A is a square-zero extension.

Lemma 1.13. *The spherical Witt ring $\mathbb{W}(A)$ can be written as a 2ω-indexed tower limit, with each term identified with a finite sum of shifts $\Sigma^j A$ and each connection map being a square-zero extension.*

Proof. The canonical map $\mathbb{W}(A) \to A$ fits into a pushout square by construction. Therefore by base change we reduce to the case where $A = \mathbb{F}_p$ and $\mathbb{W}(A) = \mathbb{S}_p$. First, one can write $\mathbb{S}_p$ as an ω-indexed limit of its Postnikov tower, with each term consisting of copies of $\Sigma^j\mathbb{Z}_p$ and each connection map being a square-zero extension. Then we can write each copy of $\mathbb{Z}_p$ as an ω-indexed limit of its p-adic tower. One conclude by combining together. □

Remark 1.14. The lemma merely implies that there exists an equivalence between $\mathbb{W}(A)$ and a 2ω-indexed limit of copies of shifts of A. It seems impossible to expect such an equivalence is canonical.

Corollary 1.15. *The spherical Witt functor $\mathbb{W}(-)$ preserves arbituary products.*

1.3. **Properties of the tilt functor.** The tilt functor behaves very crude as one can expect. It can not distingush square-zero extensions, adic elements or the information outside degree zero. More explicitly, we will have the following results

Lemma 1.16. *Let $R \to S$ be a square-zero extension. Then after applying $(-)^\flat$, it becomes an equivalence.*

Proof. Since the map $R^\flat \to S^\flat$ is a map of perfect algebras, it suffices to check the equivalence on the underlying set level. However, the underlying set functor in Perf is corepresented by $\mathbb{F}_p[t^{1/p^\infty}]$. Thus by Example 1.9 and Proposition 1.11, we have equivalences

$$R^\flat \simeq \mathrm{Hom}(\mathbb{F}_p[t^{1/p^\infty}], R^\flat) \simeq \mathrm{Map}(\mathbb{S}_p[t^{1/p^\infty}], R) \simeq \mathrm{Map}(\mathbb{S}_p[t^{1/p^\infty}], S) \simeq S^\flat$$

□

Lemma 1.17. *The tilt functor $(-)^\flat$ sends the span $\pi_0(R) \leftarrow \tau_{\geqslant 0}R \to R$ to equivalences.*

Proof. Almost by definition, since when defining the tilt functor we first take the zeroth homotopy group. □

Lemma 1.18. *Let $x \in \pi_0(R)$ be an element such that $\pi_0(R)$ is x-adic complete, then we have an equivalence $\pi_0(R)^\flat \simeq (\pi_0(R)/x)^\flat$.*

Proof. By completeness, $\pi_0(R)$ can be written as a limit of the x-adic tower of $\pi_0(R)/x$, where the connection maps are square-zero extensions. It follows from Lemma 1.17 and the fact that $(-)^\flat$ preserves limits. □

2. Lubin-Tate theories

The chromatic homotopy theory associates the chromatic height of a cohomology theory with the algebraic height of a formal group. Lubin-Tate theories allow us to attach an $\mathbb{E}_\infty$-ring to universal deformations of formal groups. We first recall some basic facts about formal groups.

2.1. Recollection of formal groups.

Definition 2.1. Let A be an ordinary ring. A **formal group law** over A is a formal series $F \in A[\![t_1, t_2]\!]$ such that

(1) $F(x, y) = F(y, x)$.
(2) $F(x, 0) = F(0, x) = x$.
(3) $F(F(x, y), z) = F(x, F(y, z))$.

We sometimes use $x +_F y$ to denote $F(x, y)$ for simplicity. An isomorphism between two formal group law F, G over A is an invertible formal series $\phi \in A[\![t]\!]$ such that $\phi(x +_F y) = \phi(x) +_G \phi(y)$.

Remark 2.2. The second requirement for a formal group law in Definition 2.1 is important. It forces the group unit to be 0. As a consequence, any formal group law must have forms

$$F(x, y) = x + y + \text{heigher terms}$$

Remark 2.3. Formal groups emerge naturally from cohomology theories. Suppose we have a complex orientable ring spectrum E, by Atiyah-Hirzrebruch spectral sequence $E^*(\mathbb{CP}^\infty) \cong E_*[\![t]\!]$. The homotopy product $m : \mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ induces

$$\begin{aligned} m^* : E_*[\![t]\!] &\to E_*[\![t_1, t_2]\!] \\ t &\mapsto F(t_1, t_2) \end{aligned}$$

which determines a formal group law on E^*. Therefore, one has a correspondence

$$\{\text{Complex oriented cohomology theory}\} \to \{\text{Formal group laws}\}$$

Not every formal group law comes from some complex oriented homotopy theory. The obstruction is Landweber exactness. However, we can always reflect a cohomology theory from those universal deformations of formal groups, as we will see later.

Let $\mathcal{FGL} : \mathrm{CRing} \to \mathcal{S}$ be the functor that assigns each ring the groupoid of formal group laws over it. This functor is actually a scheme and is represented by $\mathrm{Spec}(L)$. The ring L can be defined as follows. Let $\mathbb{Z}[a_{ij}]$ be the polynomial ring and choose $F \in \mathbb{Z}[a_{ij}][\![x, y]\!]$ as $F = \Sigma_{i,j} a_{i,j} x^i y^j$. Let $I \in \mathbb{Z}[a_{ij}]$ be the ideal generated by the coeffitients of the relations

(1) $F(x, y) - F(y, x)$.
(2) $F(x, 0) - x$.
(3) $F(F(x, y), z) - F(x, F(y, z))$.

We define $L = \mathbb{Z}[a_{ij}]/I$. One can read from the definition that we must have $\mathcal{FGL}(R) \cong \mathrm{Hom}(L, R)$.

In fact, we have a more manageable description of L

Theorem 2.4 (Lazard)**.** *The ring L is isomorphic to the graded ring $\mathbb{Z}[b_1, \cdots, b_n, \cdots]$ with b_i in degree $2i$.*

We refer to L as the **Lazard ring**. The graded ring structure here is meaningful. In fact, we have the following result.

Theorem 2.5 (Quillen)**.** *Let $\varphi : L \to \mathrm{MU}_*$ be the ring map that classifies the formal group law on MU_* determined by the complex orientation. Then $\varphi : L \xrightarrow{\simeq} \mathrm{MU}_*$ is an isomorphism of graded rings.*

Now we introduce the concept of formal groups, which reform formal group laws into geometric objects.

Definition 2.6. Let A be a ring and F be a formal group law over A. We let $\mathbb{G}_F := \mathrm{Spf} A[\![t]\!]$ be the formal scheme equipped with a multiplication $\mathbb{G}_F \times \mathbb{G}_F \to \mathbb{G}_F$ defined by

$$\begin{aligned} \mu^* : A[\![t]\!] &\to A[\![t_1, t_2]\!] \\ t &\mapsto F(t_1, t_2) \end{aligned}$$

A **formal group** over A is an abelian étale sheaf over A which is locally represented by $\mathbb{G}_F$.

Example 2.7. We use $\mathbb{G}_a$ to denote the formal group $\mathbb{G}_F$ corresponding to the formal group law $F(x, y) = x + y$. We use $\mathbb{G}_m$ to denote the formal group $\mathbb{G}_F$ corresponding to the formal group law $F(x, y) = x + y + xy$.

Now we fix a prime p and let R be an $\mathbb{F}_p$-algebra. Let $\mathbb{G}$ be a formal group over R. The Frobenius on R induces $\mathrm{Frob} : \mathbb{G} \to \mathbb{G}$. We can use this to define the height of a formal group.

Definition 2.8. Let R be an $\mathbb{F}_p$-algebra and let $\mathbb{G}$ be a formal group over R.

(1) We say $\mathbb{G}$ is at least of height n if the multiplication by p on $\mathbb{G}$ factors as

$$\begin{array}{ccc} \mathbb{G} & \xrightarrow{\mathrm{Frob}^n} & \mathbb{G} \\ & \searrow^{\cdot p} & \downarrow{\scriptstyle \phi} \\ & & \mathbb{G} \end{array}$$

(2) We say $\mathbb{G}$ is exactly of height n if it is at least of height n and ϕ is an isomorphism.
(3) We say $\mathbb{G}$ is of infinite height if it is at least of height n for every n.

Remark 2.9. In general, the height does not give a strict stratification of formal groups. That means, there exists some formal group that is at least of height n but is neither at least of height $n + 1$ nor exactly of height n. However, when R is a field, every formal group that is not of infinite height will admits a strict height n. In this case, we will use $\mathrm{ht}(\mathbb{G})$ to denote the height of $\mathbb{G}$.

Remark 2.10. When $R = k$ is a algebraic-closed field, for every height n there exists a unique formal group up to isomorphism.

The height of $\mathbb{G}_F$ can be computed at the level of formal groups. The multiplication by p on $\mathbb{G}_F$ induces a map

$$[p] : \mathrm{Spf}(R[\![t]\!]) \to \mathrm{Spf}(R[\![t]\!])$$

sending t to some formal series. Then $\mathbb{G}_F$ is at least of height n if and only if $[p](t) = h(t^{p^n})$ for some formal series h, $\mathbb{G}_F$ is exactly of height n if and only if $[p](t) = h(t^{p^n})$ and h is invertible.

Example 2.11. In Example 1.9, one can write down the formula of $[p]$ for $\mathbb{G}_a$ and $\mathbb{G}_m$ and after examing the coefficience of the formal series defined by $[p]$, one observes that $\mathrm{ht}(\mathbb{G}_a) = \infty$ and $\mathrm{ht}(\mathbb{G}_m) = 1$.

If we choose v_i to be the coefficience of t^{p^i} in the formal series of $[p]$ for the universal formal group law on MU_*, where each v_i locates in degree $2p^i - 2$, then for any formal group law (R, F), R admits a stratification by

$$(v_0 = p) \subset (v_0, v_1) \subset (v_0, v_1, v_2) \subset \cdots \subset R$$

The quotient $R/(v_0, \cdots, v_n)$ cuts out the $\mathrm{ht} \leqslant n$ part of the formal group law. We define $\mathfrak{m}_n = (v_0, \cdots, v_n)$ and refer to it as the n-**th Landweber ideal** of R.

2.2. **Lubin-Tate functors.** We now introduce the Lubin-Tate functor and compare it with the spherical Witt functor. We begin with some notations.

Definition 2.12. We say an $\mathbb{E}_\infty$-ring R is complex periodic if the followings are satisfied

(1) R is complex orientable.
(2) $\pi_2(R)$ is a rank 1 projective $\pi_0(R)$-module.
(3) The multiplication $\pi_2(R) \otimes_{\pi_0(R)} \pi_n(R) \to \pi_{n+2}(R)$ is an equivalence.

For a complex periodic $\mathbb{E}_\infty$-ring R, we can choose a unit $\mu \in \pi_{-2}(R)$ and press $v_0, v_1, \cdots, v_n$ onto degree 0 by multiplying a power of μ.

Definition 2.13. Let R be an even $\mathbb{E}_\infty$-ring. $\pi_*(R)$ is stratified with Landweber ideals $\mathfrak{m}_i$. One can lift $\pi_*(R)/\mathfrak{m}_n$ to a spectrum $R//\mathfrak{m}_n$, which is defined as

$$R//\mathfrak{m}_n := R/v_0 \otimes \cdots \otimes R/v_n$$

By evenness each R/v_i admits an $\mathbb{E}_1$-structure. Therefore, $R//\mathfrak{m}_n$ gives $\pi_*(R)/\mathfrak{m}_n$ a $\mathbb{E}_1$-lift. We can now introduce the Lubin-Tate functor.

Theorem 2.14 ([Lur17a], Theorem 5.0.2)**.** *Let $\mathcal{FG}^{=n}$ be the category of pairs $(A, \mathbb{G})$ consisting of a perfect $\mathbb{F}_p$-algebra A and a formal group $\mathbb{G}$ of height n. A morphism $\varphi : (A, \mathbb{G}_0) \to (B, \mathbb{G}_1)$ is a pair (f, α), where $f : A \to B$ is a map of rings and $\alpha : f^*\mathbb{G}_0 \xrightarrow{\simeq} \mathbb{G}_1$ is an equivalence. Then there is a functor*

$$E(-;-) : \mathcal{FG}^{=n} \to \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$$

such that

(1) $E(-;-)$ is fully faithful.
(2) For every $(A, \mathbb{G}) \in \mathcal{FG}^{=n}$, $E(A; \mathbb{G})$ is $K(n)$-local and even.
(3) The elements $v_0, \cdots, v_{n-1}$ are regular.
(4) The essential image of $E(-;-)$ consists of $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ such that $R//\mathfrak{m}_{n-1}$ is even and $\pi_0(R)/\mathfrak{m}_{n-1}$ is perfect.

(5) *The classical Quillen's formal group over* $\pi_0 E(A;\mathbb{G})$ *is identified with the universal deformation of* $\mathbb{G}$.
(6) *We have an equivalence*

$$\pi_* E(A;\mathbb{G}) \cong W(A)[\![v_1,\cdots,v_{n-1}]\!][\mu^{\pm 1}]$$

where $v_1,\cdots,v_{n-1}$ *locates in degree* 0 *and* μ *is in degree* 2.

Therefore, if we fix a perfect $\mathbb{F}_p$-field k and a given formal group $\mathbb{G}_0$, the functor is simplified to

$$E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}^{\wedge}_{E(k)}$$

which sends $f : k \to A$ to $E(A; f^*\mathbb{G}_0)$.

Remark 2.15. When $A = k$ is a perfect field, then as a spectrum, $E(-,-)$ does not rely on the choice of formal groups, see [LP22].

Now we compare the functor $E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ with $\mathbb{W}(-) : \mathrm{Perf} \to \mathrm{CAlg}(\mathrm{Sp}_p)$.

Let $\mathcal{C}$ be a p-complete presentable symmetric monoidal stable ∞-category. The unit functor $\mathbb{S}_p \to \mathcal{C}$ in $\mathrm{CMon}(\mathrm{Pr}^{\mathrm{L}}_{st})$ gives an adjunction

$$\iota^* : \mathrm{CAlg}(\mathrm{Sp}_p) \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : \iota_*$$

Composing this with the Witt-tilt adjunction, we have

$$\mathbb{W}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

where we still use $\mathbb{W}(-)$ and $(-)^\flat$ to denote this pair. The unit $\mathbb{1}_{\mathcal{C}} \in \mathrm{CAlg}(\mathcal{C})$ is sending to some $B \in \mathrm{Perf}$. We obtain an adjunction

$$\mathbb{W}_B(-) : \mathrm{Perf}_B \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

which from left to right sends A to $\mathbb{1}_{\mathcal{C}} \otimes_{\iota^*\mathbb{W}(B)} \iota^*\mathbb{W}(A)$.

Now we take $\mathcal{C} = \mathrm{Mod}^{\wedge}_{E(k)}$ to be the category of $T(n)$-local $E(k)$-module. Apply the above construction, we obtain

$$\mathbb{W}_B(-) : \mathrm{Perf}_B \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

for $B = E(k)^\flat$.

Lemma 2.16. *For any* $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, *we have*

$$R^\flat \simeq \pi_0(R)^\flat \simeq (\pi_0 R/\mathfrak{m}_{n-1})^\flat$$

In particular, we have $E(k)^\flat \simeq k^\flat \simeq k$ *and* $E(-)^\flat \simeq \mathrm{id}$.

Proof. This follows from the discussion in Section 1.3, the fact that ι_* preserves square-zero extensions, and the equivalence between $T(n)$-local and $\mathfrak{m}_n$-complete ([HS99, Proposition 7.10]). □

Therefore, the above adjunction has the form

$$\mathbb{W}_k(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

Since $E(-)^\flat \simeq \mathrm{id}$, precomposing with the unit map of the adjunction allows us to obtain a natural transformation

$$\mathbb{W}_k(-) \to E(-)$$

One can show that this map is an equivalence by reading the homotopy groups, since we have an equivalence

$$\mathbb{W}_k(A) \simeq \mathbb{W}(A) \otimes_{\mathbb{W}(k)} E(k) \simeq E(A)$$

To conclude, we have the following consequence.

Theorem 2.17. *Fix a perfect field k of character p and a height n, then there is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

such that

1. *$E(-)$ is fully faithful.*
2. *$(-)^\flat \simeq \pi_0(-)^\flat \simeq (\pi_0(-)/\mathfrak{m}_{n-1})^\flat$.*
3. *$\pi_* E(A) \cong W(A)[\![v_1, \cdots, v_{n-1}]\!][\mu^{\pm 1}]$. In particular, $p = v_0, v_1, \cdots, v_{n-1}$ is a regular sequence and $E(A)$ is even.*
4. *The essential image of $E(-)$ consists of $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$ such that $R//\mathfrak{m}_{n-1}$ is even and $\pi_0(R)/\mathfrak{m}_{n-1}$ is perfect.*

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LECTURE 3

TYPED BY AOWEN FAN

Through this talk, we will discuss the power opertations and their relationship with the deformation theory. Our main aim is to prove the cofreeness of the Lubin-Tate theory. We will cover the contents of [BSY22, Section 3]. We also give a more handful proof of this main theorem at the end of the talk.

1. Power operations

Let E be an $\mathbb{E}_\infty$-ring spectrum. We work in the world of E-algebras. The power operations are some natural transformations $\pi_i(-) \to \pi_j(-)$ of functors $\mathrm{hAlg}_E \to \mathrm{Sets}$. In hAlg_E, π_i is corepresented by $E\{t\} = \mathbb{P}(\Sigma^i E) \in \mathrm{hAlg}_E$, where $\mathbb{P}(-) = \bigoplus_{m\geqslant 0} \mathbb{P}_m(-)$ is the free commutative E-Alg monad on hMod_E. By Yoneda, one has the following bijection between sets.

$$\begin{array}{ccc} \{\text{Power operations } \theta : \pi_i(A) \to \pi_j(A)\} & \longleftrightarrow & \pi_j(\mathbb{P}(\Sigma^i E)) \\ \uparrow & & \uparrow \\ \{\text{Additive power operations } \theta : \pi_i(A) \to \pi_j(A)\} & \longleftrightarrow & \{\text{Primitive elements in } \pi_j(\mathbb{P}(\Sigma^i E))\} \end{array}$$

The examples of power operations include the Steenrod operation and Adams operations on p-complete K-theory. Through this talk we focus on $E = E(k)$ for a fixed prime p and a height n. In [Rez09], Rezk constructed an algebraic approximation of $\mathbb{P}$, which is a monad on Mod_{E_*} denoted by $\mathbb{T}$.

The construction of Rezk can be concluded as follows. We consider the equivalence induced by π_*

$$\mathrm{hMod}_E^{ff} \xrightarrow{\simeq} \mathrm{Mod}_{E_*}^{ff}$$

If we restrict to $K(n)$-local finite free E-algebras, then $\mathbb{P}_m$ becomes a submonad. One can define the desired $\mathbb{T}_m$ as the left Kan extension

$$\begin{array}{ccccc} \mathrm{hMod}_E^{ff} & \hookrightarrow & \mathrm{hMod}_E & \xrightarrow{\pi_* L_{K(n)} \mathbb{P}_m} & \mathrm{Mod}_{E_*} \\ \downarrow \simeq & & \downarrow \pi_* & \nearrow_{\mathbb{T}_m} & \\ \mathrm{Mod}_{E_*}^{ff} & \hookrightarrow & \mathrm{Mod}_{E_*} & & \end{array}$$

2. Main theorem

We consider the forgetful functor $U_{\mathbb{T}}$ from $\mathbb{T}$-algebras to E_0-algebras, which admits both a left adjoint $F_{\mathbb{T}}$ and a right adjoint $W_{\mathbb{T}}$.

$$\mathrm{Alg}_{\mathbb{T}} \underset{W_{\mathbb{T}}}{\overset{F_{\mathbb{T}}}{\underset{\longrightarrow}{\overset{\longleftarrow}{U_{\mathbb{T}}}}}} \mathrm{CRing}_{E_0}$$

Applying the above adjunction to the height-1 case and considering the classical adjunction

$$(-/p)^\sharp : \mathrm{CRing} \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \mathrm{incl}$$

We obtain an adjunction

$$(U(-)/p)^\sharp : \mathrm{CRing}_\delta \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \pi_0 E(-)$$

Our main theorem generalizes this observation to arbituary height.

Theorem 2.1. *We fix a height n and let $\mathfrak{m} = \mathfrak{m}_n$ be the Landweber ideal. Then there is an adjunction*

$$(U(-)/\mathfrak{m})^\sharp : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \pi_0 E(-)$$

where the right adjoint $\pi_0 E(-)$ is fully faithful.

Definition 2.2. Let $\mathbb{1}$ be the symmetric unit in a stable symmetric monoidal category $\mathcal{C}$. To each $P \in \pi_0(\mathbb{1}_{h\Sigma_r})$, we can associate a power operation of weight r, which takes $x \in \pi_0(R)$ to

$$\mathbb{1} \xrightarrow{P} \mathbb{1}_{h\Sigma_r} \to \oplus \mathbb{1}_{h\Sigma_r} \simeq \mathbb{1}\{t\} \to R$$

This induces a map $P_r : \pi_0 R \to [B\Sigma_r, R]$. We call it the total power operation of weight r.

Let A be a perfect k-algebra. The underlying set of $W_{\mathbb{T}}(A)$ can be identified with the set of A-valued functions on E_0-mod of power operations. Let $ev_A : A \to W_{\mathbb{T}} U_{\mathbb{T}}$. Consider the following diagram

$$\begin{array}{ccc} & W_{\mathbb{T}} U_{\mathbb{T}} \pi_0 E(A) & \\ {\scriptstyle ev_A}\nearrow & & \searrow{\scriptstyle W_{\mathbb{T}}(-/\mathfrak{m})} \\ \pi_0 E(A) & \xrightarrow{\overline{ev}_A} & W_{\mathbb{T}}(A) \end{array}$$

we will prove that $\overline{ev}_A$ is an equivalence for every $A \in \mathrm{Perf}_k$, then everything follows immediately.

3. The proof

3.1. **The Witt filtration.** For $\lambda \in \mathbb{T}(E_0)$, we have an evaluation map $\lambda^* : W_{\mathbb{T}}(A) \to A$. We define an ideal

$$W_{\mathbb{T}}^{\geqslant r}(A) = \{v \in W_{\mathbb{T}}(A) | \lambda^* v = 0 \text{ for any power operation } \lambda \text{ with weight less than } p^r\}$$

Definition 3.1. We let $W_{\mathbb{T}}^{\leqslant r}(A) = W_{\mathbb{T}}(A)/W_{\mathbb{T}}^{\geqslant r+1}(A)$. This defines a filtration which we call the **Witt filtration**.

$$W_{\mathbb{T}}(A) \to \cdots \to W_{\mathbb{T}}^{\leqslant r}(A) \to \cdots \to W_{\mathbb{T}}^{\leqslant 1}(A) \to W_{\mathbb{T}}^{\leqslant 0}(A) = A$$

Strickland shows that we have an equivalence of underlying sets

$$W_{\mathbb{T}}(A) \cong \mathrm{Hom}_{E_0}(\mathbb{T}(E_0), A) \cong \prod_{r \geqslant 0} \overline{d}(r)$$

where $\overline{d}(r)$ denotes the Gaussian polynomial coefficience.

Proposition 3.2 ([BSY22]; Proposition 3.27)**.** *For any $A \in \mathrm{Alg}_{E_0}$, we have an equivalence of graded pieces*

$$W_{\mathbb{T}}^{=r}(A) = W_{\mathbb{T}}^{\geqslant r}(A)/W_{\mathbb{T}}^{\geqslant r+1}(A)$$

which fits into the commutative diagram

$$\begin{array}{ccc} W^{\geqslant r}(A) & \longrightarrow & W^{=r}(A) \\ \cong\downarrow & & \downarrow \\ \prod_{i \geqslant r} \overline{d}(i) & \longrightarrow & \overline{d}(r) \end{array}$$

The A-mod structure is not obvious. Instead, one can show that the A-operation is given by $[a]v = a^{p^r}v$. As a result, $W_{\mathbb{T}}^{=r}(A)$ becomes a free A-module of rank $\overline{d}(r)$.

One can use this to form the base-change for reduction. We have the following consequence which allows us to reduce to k.

Theorem 3.3. *If $\overline{ev}_k : \pi_0 E(k) \to W_{\mathbb{T}}(k)$ is an isomorphism, then $\overline{ev}_A$ is an isomrophism for every $A \in \mathrm{Perf}_k$.*

Proof. We have the following commutative diagram, where the vertical arrows are completion maps.

$$\begin{array}{ccc} W(A) \otimes_{W(k)} \pi_0 E(k) & \longrightarrow & W(A) \otimes_{W(k)} W_{\mathbb{T}}(k) \\ \wedge\downarrow & & \downarrow \\ \pi_0 E(A) & \longrightarrow & W_{\mathbb{T}}(A) \end{array}$$

It follows from the observation that the $\mathfrak{m}$-adic topology and W-topology coincide. □

3.2. **Surjectivity.** We want to show the following.

Proposition 3.4. *For every weight r, the mod $\mathfrak{m}$ total power operation $\overline{\tau}_r : \pi_0 E(k) \to E^0(B\Sigma_{p^r})/(I_{tr}, \mathfrak{m})$ is surjective.*

In order to prove the surjectivity, we need to use the Moduli interpretion of power operations.

We consider the moduli problem. We consider the functor LT from complete local rings to groupoids, sends R to deformations of G_0 over k. This is correpresented by a Lubin Tate ring $\mathcal{O}_{\mathrm{LT}}$, which non-cannonically isomorphic to $W(k)[\![u_1, \cdots, u_{n-1}]\!]$.

Theorem 3.5 ([AHS04]; Proposition 12.12)**.** *Let E be the Lubin-Tate theory attached to a height n formal group $\mathbb{G}_0$ over a perfect field k. Let $\varphi^r : \mathbb{G}_0 \to \mathbb{G}_0^{(r)}$ be the r-fold Frobenius isogeny. Then there is an isomorphism of schemes*

$$\mathrm{Def}(\varphi^r) \cong \mathrm{Spf}(E^0(B\Sigma_{p^r})/I_{tr})$$

Therefore, in order to prove Proposition 3.4, it suffices to show we have a close immersion

$$\mathrm{Def}(\varphi^r) \xrightarrow{(\sigma.\tau)} \mathrm{Def}(\mathbb{G}_0) \times_{\mathrm{Spf}W(k)} \mathrm{Def}(\mathbb{G}_0^{(r)})$$

which follows from a more general fact in algebraic geometry.

Proposition 3.6. *Let $q_0 : \mathbb{G}_0 \to \mathbb{G}_0'$ an isogeny of formal groups over l. Then*

$$\mathrm{Def}(q_0) \xrightarrow{(\sigma.\tau)} \mathrm{Def}(\mathbb{G}_0) \times_{\mathrm{Spf}W(k)} \mathrm{Def}(\mathbb{G}_0')$$

is a closed immersion.

3.3. **Complete the proof.** We can complete the proof that $\overline{ev}_A : A \to W_{\mathbb{T}}U_{\mathbb{T}}A$ is an isomorphism.

Definition 3.7. A p-derivation is $\theta \in \mathbb{T}(E_o)_p$ such that for any $\mathbb{T}$-algebra B and $x, y \in B$, $\theta(x+y) = \theta(x) + \theta(y) + (x^p + y^p - (x+y)^p)/p$.

Using induction for height and regard the whole paper as inductive, one can show the following

Lemma 3.8 ([BSY22]; Proposition 3.45)**.** *For any nonezero $y \in E_0$, there is an additive power operation Q together with a p-derivation θ such that $\theta^i Q(y)$ is nonzero.*

Therefore, one can conclude the results by assemble the consequences for $\overline{\tau}_r : \pi_0 E(k) \to E^0(B\Sigma_{p^r})/(I_{tr}, \mathfrak{m})$.

Theorem 3.9. *The evaluation map $\overline{ev}_k : E_0(k) \to W_{\mathbb{T}}(k)$ is an isomorphism.*

Proof. The evaluation map can be interpreted as evaluation functions in power operations. The injective part comes from Lemma 3.8. The surjective part is from Proposition 3.4 and additional length arguments. □

3.4. **Akhil's proof.** The proof of [BSY22] is somehow technical. Akhil gives a more concise proof.

Let Ψ be the power operation that corresponds to the isogeny $[p]$ through Strickland's correspondence. This implies that Ψ is central in additive power operations. We say a $\mathbb{T}$-algebra R is perfect if $\Psi : R \to R$ is an isomorphism. We can obtain a colimit perfection of $R \in \mathrm{Alg}_{\mathbb{T}}$.

$$R^{\mathrm{Perf}} = \mathrm{colim}(R \xrightarrow{\Psi} R \xrightarrow{\Psi} R \xrightarrow{\Psi} \cdots)$$

We have an adjunction

$$(\hat{-})^{\mathrm{Perf}} : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge} : \mathrm{forget}$$

It suffices to prove that we have an equivalence

$$(-)/\mathfrak{m} : \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge} \xrightarrow{\simeq} \mathrm{Perf}_k$$

The essentially full is clear. To see that the functor is fully faithful, the key observation is the following

Lemma 3.10. *The functor $(\hat{-})^{\mathrm{Perf}} : \mathrm{Alg}_{\mathbb{T}} \to \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge}$ sends R to $E_0(A)$ for some perfect k-algebra A.*

Proof idea. We can define $A := (\hat{R}//\mathfrak{m})^{\mathrm{Perf}}$ and check that $\hat{R}^{\mathrm{Perf}} \simeq E_0(A)$ as $\mathbb{T}$-algebras. □

Therefore, every perfect $\mathfrak{m}$-complete $\mathbb{T}$-algebra is of the form $E_0(A)$. By formal étaleness we have

$$\mathrm{Alg}_{\mathbb{T}}(E_0(A_1), E_0(A_2)) \subset \mathrm{Alg}_{E_0}(E_0(A_1), E_0(A_2)) = \mathrm{Alg}_k(A_1, A_2)$$

which admits a section sending $A_1 \to A_2$ to $E_0(A_1) \to E_0(A_2)$.

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LECTURE 4

TYPED BY AOWEN FAN

In this talk, we collect main results of Section 4 of [BSY22], which will offer a main tool in our next talk. We discuss the nilpotence detection in a locally rigid category. Moreover, we will apply nilpotence detecting techniques to Lubin-Tate-tilt adjunction and give a criterion for a ring to be perfect of Krull dimension 0.

1. Locally rigid categories

Definition 1.1. We use $\mathrm{Pr}_L^{\mathrm{rig}}$ to denote the category of locally rigid compactly generated categories, with morphisms the strongly continuous functors.

The following lemmas are fundamental.

Lemma 1.2. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$, then*

(1) $c \in \mathcal{C}^\omega$ implies $c^\vee \in \mathcal{C}^\omega$.
(2) Suppose we have $a \in \mathcal{C}$ such that $a \otimes c = 0$ for any $c \in \mathcal{C}^\omega$, then $a = 0$.

Lemma 1.3. *Consider $\mathrm{Mod}_B(\mathcal{C})$ for $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$ and $B \in \mathrm{CAlg}(\mathcal{C})$. Then*

(1) $\mathrm{Mod}_B(\mathcal{C})$ is in $\mathrm{Pr}_L^{\mathrm{rig}}$.
(2) $- \otimes B : \mathcal{C} \to \mathrm{Mod}_B(\mathcal{C})$ is strongly continuous.

2. Nilpotence and detecting nilpotence

Definition 2.1. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$. Suppose $f : a \to b$ is a morphism in $\mathcal{C}$ and $e \in \mathcal{C}$.

(1) We say that f is nilpotent at e, if there is some $m \in \mathbb{N}$ such that $f^{\otimes m} \otimes \mathrm{id} : a^{\otimes m} \otimes e \to b^{\otimes m} \otimes e$ is null.
(2) We say f is nilpotent if it is nilpotent at $\mathbb{1}_{\mathcal{C}}$.
(3) We say f is locally nilpotent if it is nilpotent at every compact object.

The following lemma is immediate.

Lemma 2.2. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$.*

(1) Suppose we have $f : c \to a$ with c compact, then f is locally nilpotent if and only if f is nilpotent.
(2) Suppose d is dualizable. Then $f : d \otimes a \to b$ is nilpotent at e if and only if $f^\vee : a \to b \otimes d^\vee$ is nilpotente at e.
(3) Suppose d is dualizable, then f is nilpotent at d if and only if $f \otimes d$ is nilpotent.

Definition 2.3. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$, $T \in \mathcal{C}$ and $c \in \mathcal{C}^\omega$.

(1) We say T detects nilpotence at c if for every map $f : c \to a$, f is nilpotence at T if and only if it is nilpotence.
(2) We say T detects nilpotence if it detects nilpotence at every compact object.

In reality, we will deal with the weak rings, that means, objects T in $\mathcal{C}$ that with a unit map $\mu_T : \mathbb{1}_{\mathcal{C}} \to T$ such that $\mu_T \otimes T$ admits a retract.

Lemma 2.4. *Suppose T_1, T_2 are two weak rings. Then T_1, T_2 detect nilpotence imply $T_1 \otimes T_2$ detects nilpotence.*

Proposition 2.5. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$, $T \otimes -$ is conservative, then T detects nilpotence.*

Example 2.6. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ and $e \xrightarrow{x} \mathbb{1}_{\mathcal{C}}$ with $e \in \mathrm{Pic}(\mathcal{C})$. Then $\mathbb{1}_{\mathcal{C}}[x^{-1}] \oplus \mathbb{1}_{\mathcal{C}}/x$ detects nilpotence.

Example 2.7. Let $R \in \mathrm{CAlg}_{E(k)}^{\wedge}$. Since each v_i is locally nilpotence, then $R//\mathfrak{m} = R/v_1 \otimes_R \cdots \otimes_R R/v_{n-1}$ detects nilpotence in $\mathrm{Mod}_R^{\wedge}$.

Proposition 2.8. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$.*

1. *Suppose that $T \in \mathcal{C}$ is a weak ring, then T detects nilpotence, then $B \otimes T$ detects nilpotence in $\mathrm{Mod}_B(\mathcal{C})$.*
2. *Filtered colimits of nilpotence detecting weak rings are nilpotence detecting weak rings.*

The key results for nilpotence detecting theory is the following.

Theorem 2.9. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$. Then nilpotence detecting morphisms in $\mathrm{CAlg}(\mathcal{C})$ form a weakly saturated collection.*

Recall that we say $f : C \to D$ has right lifting property with respect to $X \in \mathcal{C}$ if there is always a solution

$$\begin{array}{ccc} C & \longrightarrow & X \\ {\scriptstyle f}\downarrow & \nearrow & \\ D & & \end{array}$$

In this case, we denote $f \perp X$.

By Quillen's small objects argument, we can prove the following.

Theorem 2.10. *In $\mathrm{CAlg}(\mathcal{C})$:*

1. *$\mathbb{1}[t] \to \mathbb{1}[t^{\pm 1}] \times \mathbb{1}$ detects nilpotence.*
2. *$\mathbb{1}[t^{1/p^\infty}] \to \mathbb{1}[t^{\pm 1/p^\infty}] \times \mathbb{1}$ detects nilpotence.*

3. Apply to perfect rings of Krull dimension 0

We have the following criterion for Krull dimension 0 commutative rings.

Proposition 3.1. *Let B be an ordinary commutative ring, then the followings are equivalent.*

1. *B is reduced of Krull dimension 0*
2. *We have*

$$(\mathbb{Z}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{Z}[t^{\pm 1}] \times \mathbb{Z}) \perp B$$

Similarly, if we restrict ourselves to perfect rings, we have

Proposition 3.2. *Let A be a perfect ring, then the followings are equivalent.*

1. *A is reduced of Krull dimension 0*
2. *We have*

$$(\mathbb{F}_p[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{F}_p[t^{\pm 1}] \times \mathbb{F}_p) \perp B$$

Corollary 3.3. *Suppose B is an $\mathbb{F}_p$-algebra such that $B^\flat$ is of Krull dimension 0, then the natural map $B^\flat \to B$ is injective.*

One can apply the Lubin-Tate-tilt adjunction. We have

Corollary 3.4. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ be p-complete. In $\mathrm{CAlg}(\mathcal{C})$, consider*

$$f : \mathbb{1}_{\mathcal{C}}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}}$$

Then $f \perp R$ if and only if $R^\flat \in \mathrm{Perf}_p$ is of Krull demension 0.

Theorem 3.5. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ be p-complete. Recall that we have a spherical-tilt adjunction*

$$\mathbb{W}_{\mathcal{C}}(-) : \mathrm{Perf}_p \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

Let $f : A \to B$ be a morphism in Perf_p. Then the following are equivalent

(1) $\mathbb{W}_{\mathcal{C}}(B) \in \mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(A)}$ is conservative
(2) The map $\mathbb{W}_{\mathcal{C}}(A) \to \mathbb{W}_{\mathcal{C}}(B)$ detects nilpotence.
(3) The base-change functor $\mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(A)} \to \mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(B)}$ is nil-conservative.
(4) $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective.

Conclusions

(1) Detecting nilpotence means that the corresponding maps between Spec are surjective.
(2) $(\mathbb{1}_{\mathcal{C}}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}})R$ means R is of Krull dimension 0. By small objects argument, this gives a cover

$$\mathrm{Spec}(\prod_{A\to k} k) \to \mathrm{Spec}(A)$$

We will show that under some conditions the former can be written as the spectrum of a Lubin-Tate theory in our next talk.

References

[BSY22] Robert Burklund, Tomer M Schlank, and Allen Yuan. "The chromatic nullstellensatz". In: *arXiv preprint arXiv:2207.09929* (2022) (cit. on p. 1).

LECTURE 5

AOWEN FAN

In this talk, we will prove the main theorems in the Section 5 and Section 6 of [BSY22]. We first prove that for any $T(n)$-local algebra R there always exists a perfect algebra A of Krull dimension 0 and a nilpotence-detecting map $R \to E(A)$. As a consequence, we can map any $T(n)$-local algebra to the Lubin-Tate theory of some algebraic closed field L.

In addition, we define the Nullstellensatzian objects in a presentable category. We show the Nullstellensatzian theorem. That is, the Nullstellensatzian objects in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ are exactly the Lubin-Tate theories attached to algebraic closed fields. At the end of this talk, we will build an analogy of Zariski spectra on $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$.

1. Recollection

We briefly recollect some key results in the previous talks. Recall that we introduced the Lubin-Tate functor. More explicitly, we have:

Proposition 1.1. *Let k be a perfect field of character p and we fix a formal group $\mathbb{G}$ of height n. There is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

which we refer to as the Lubin-Tate-tilt adjunction, such that:

1. *$E(-)$ is fully faithful.*
2. *The essential image of $E(-)$ consists of the rings R such that R is even periodic, $\pi_0(R)/\mathfrak{m}$ is perfect and $R//\mathfrak{m}$ is even.*

We have shown that the Lubin-Tate theory is cofree. That is, there is an adjunction

$$(U_{\mathbb{T}}(-)/\mathfrak{m})^{\flat} : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{Perf}_k : \pi_0 E(-)$$

such that $\pi_0 E(-)$ becomes a fully faithful right adjoint.

We proved the cofreeness by showing that the evaluation maps

$$\overline{ev}_A : \pi_0 E(A) \to W_{\mathbb{T}}(\pi_0 E(A)/\mathfrak{m})$$

become isomorphisms. As a result, we have

Corollary 1.2. *Suppose we have a map $f : \pi_0 E(A) \to \pi_0 R$. Then f is injective if and only if $f/\mathfrak{m}$ is injective.*

Proof. We can approximate f by the evaluation map. After applying $\overline{ev}_A$, we obtain a commutative diagram

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$$\begin{array}{ccc} \pi_0 E(A) & \xrightarrow{\overline{ev}_A} & W_{\mathbb{T}}(\pi_0 E(A)/\mathfrak{m}) \\ {\scriptstyle f}\downarrow & & \downarrow \\ \pi_0 R & \longrightarrow & W_{\mathbb{T}}(\pi_0 R/\mathfrak{m}) \end{array}$$

The above arrow is an isomorphism and the right arrow is an injection, which forces f to be an injection. Here we use the fact that $W_{\mathbb{T}}(-)$ preserves injections. □

Throughout the last talk, we introduced the nilpotence detecting map. We have the following notation.

Definition 1.3. Let $f : C \to D$ be a morphism in $\mathcal{C}$ and R be an object. We say that $f : C \to D$ has right lifting property with respect to R and denote $f \perp R$ if we always have a lifting

$$\begin{array}{ccc} C & \longrightarrow & R \\ {\scriptstyle f}\downarrow & \nearrow & \\ D & & \end{array}$$

By Quillen's small object argument, we can show the following.

Proposition 1.4. *Let $S = \{f_s : C_s \to D_s\}$ be a family in $\mathcal{C}$ consisting of nilpotence-detecting morphisms. For any $R \in \mathcal{C}$, there exists an extention $R \to R'$ such that*

(1) $R \to R'$ detects nilpotence.
(2) For any $f_s \in S$, $f_s \perp R'$.

The spirit of the nilpotence detection is that we can convert the properties of objects to the properties of morphisms. We have a correspondence

$$\{\text{properties of the objects}\} \Leftrightarrow \{\text{family of morphisms}\}$$

such that an object R satisfies these properties if and only if the family of morphisms has right lifting property with respect to R. For example, we can use this principle to detect rings of Krull dimension 0.

Proposition 1.5. *Let $R \in \mathrm{CRing}$. R is of Krull dimension 0 if and only if $f \perp R$, where f is defined as*

$$f : \mathbb{Z}[t] \xrightarrow{t \mapsto (t,0)} \mathbb{Z}[t^{\pm 1}] \times \mathbb{Z}$$

Variant 1.6. We have the followings.

(1) Let $A \in \mathrm{Perf}_k$ be a perfect $\mathbb{F}_p$ algebra. A is of Krull dimension 0 if and only if $f' \perp R$, where f' is defined as

$$f' : \mathbb{F}_p[t] \xrightarrow{t \mapsto (t,0)} \mathbb{F}_p[t^{\pm 1}] \times \mathbb{F}_p$$

(2) Let $\mathcal{C} \in \mathrm{CMon}(\mathrm{Pr}_{st})$ be a p-complete presentable symmetric monoidal stable category. We have a Witt-tilt adjunction

$$\mathbb{W}_{\mathcal{C}}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

$R^\flat$ is of Krull dimension 0 if and only if $f'' \perp R$, where f'' is defined as

$$f' : \mathbb{1}_{\mathcal{C}}[t] \xrightarrow{t \mapsto (t,0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}}$$

We will see that the above spirit is the main ingredient for the proof of the first theorem.

2. Proof of Theorem D

We first prove the Theorem D of [BSY22].

Theorem 2.1. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then there exists a nilpotence-detecting morphism $R \to E(A)$ such that A is reduced and of Krull dimension 0.*

Once we obtain this result, the following is immediate.

Corollary 2.2 ([BSY22]; Theorem D)**.** *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then there exists a morphism $R \to E(L)$, where L is an algebraic closed field.*

We begin with the simplified case of height 0.

2.1. **The case of height 0.** Recall that when the height equals 0, everything is of character 0 and "perfect" means "reduced". In this case, $\pi_* E(A) = A[u^{\pm 1}]$ where u is a generator in degree 2.

Note that $\mathbb{Q} \to E(\mathbb{Q})$ admits a retraction. Therefore, the base change functor $- \otimes_{\mathbb{Q}} E(\mathbb{Q})$ is conservative, which implies that $\mathbb{Q} \to E(\mathbb{Q})$ detects nilpotence. As a consequence, it suffices to show

Theorem 2.3. *Let $R \in \mathrm{CAlg}^{\wedge}_{E(\mathbb{Q})}$. Then there exists a nilpotence-detecting morphism $R \to E(A)$ such that A is reduced and of Krull dimension 0.*

The proof idea is to convert the properties of rings to a family of morphisms. We want to find an extension $R \to R'$ such that $R' \simeq E(A)$. Therefore, R' should at least satisfy the following properties:

(1) R' is even.
(2) $\pi_0 R'$ is reduced and is of Krull dimension 0.

The following proposition points to the morphisms that detect these properties.

Proposition 2.4. *We define $f_0 : E(k)[t] \to E(k)[t^{\pm 1}] \times E(k)$ and $h_0 : E(k)\{z^1\} \to E(k)$. Let $S_0 = \{f_0, h_0\}$ be the family of morphisms in $\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})$ and let R be an object. Then $S_0 \perp R$ if and only if $\pi_0(R)$ is reduced of Krull dimension 0 and $\pi_*(R) = \pi_0(R)[u^{\pm 1}]$.*

Proof. By definition, it is clear that $h_0 \perp R$ if and only if $\pi_1(R) = 0$. It remains to show that $f_0 \perp R$ if and only $\pi_0(R)$ is reduced of Krull dimension 0. This is because $f_0 \perp R$ if and only if

$$\pi_0(\mathrm{Map}_{\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})}(E(k)[t^{\pm 1}] \times E(k), R)) \to \pi_0(\mathrm{Map}_{\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})}(E(k)[t], R))$$

is surjective. By freeness of $E(k)[t]$, this is equivalent to say that

$$\mathrm{Map}(\mathbb{Z}[x^{\pm 1}] \times \mathbb{Z}, \pi_0(R)) \to \mathrm{Map}(\mathbb{Z}[x], \pi_0(R)).$$

is surjective. It concludes from Variant 1.6.

□

Proposition 2.5. *The maps $\{f_0, h_0\}$ defined above detect nilpotence.*

Therefore, one can use the small object argument to conclude that one can always extend R to some R', such that R' is even and $\pi_0(R')$ is reduced of Krull dimension 0.

Unfortunately, this does not imply $R' = E(A)$. To complete the proof, we need to map R' to its residue fields. For $x \in \mathrm{Spec}(\pi_0(R'))$, let F_x denote the residue field of $\pi_0(R')$ at x. We can construct ring spectra R'_x such that $\pi_*(R'_x) \cong \pi_0(R'_x)[u^{\pm 1}] \cong F_x[u^{\pm 1}]$. We claim $R'_x \simeq E(F_x)$. Then we have a map

$$R' \to \prod_{x \in \mathrm{Spec}(\pi_0 R')} E(F_x) \simeq E(\prod_{x \in \mathrm{Spec}(\pi_0 R')} F_x)$$

which detects nilpotence since it induces a surjection on $\mathrm{Spec}(\pi_0(-))$. And $\prod_{x \in \mathrm{Spec}(\pi_0 R')} F_x$ is reduced of Krull dimension 0.

To show that $R'_x \simeq E(F_x)$, first note that $E(F_x) \simeq E(k) \otimes_k F_x$. By the ring structure, it suffices to construct a map $F_x \to R'_x$ such that it induces an isomorphism on π_0. This can always be done, since one can consider the Postnikov tower of R'_x and extend $F_x \simeq \tau_{\leqslant 0} R'_x$ along the square-zero extensions.

2.2. **The case of positive height.** From now on, we will focus on positive height. We fix a perfect field k of characteristic p and a formal group $\mathbb{G}$ of height n. We first note that in this case, $\mathbb{S}_{T(n)} \to E(k)$ detects nilpotence by [DHS88]. It suffices to show:

Theorem 2.6. *Let k be a perfect field of characteristic p, and let $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$. Then there exists a perfect k-algebra A of Krull dimension 0 and a nilpotence detecting map $R \to E(A)$.*

The proof idea is similar. Suppose we have an extension $R \to R'$ such that $R' \simeq E(A)$ for some A of Krull dimension 0, we want to associate the properties of R' with a family of morphisms. The following lemma allows us to summarize these properties.

Lemma 2.7. *Let $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$. $R \simeq E(A)$ for some A of Krull dimension 0 if and only if the following properties are satisfied.*

(1) $R^\flat$ is of Krull dimension 0.
(2) The colocalization map $\pi_0 E(R^\flat) \to \pi_0 R$ is surjective.
(3) $\pi_1(R) = 0$.

Proof. The only if part is obvious, so we prove the if part. First, notice that the condition (1) implies that $\pi_0 E(R^\flat) \to \pi_0 R$ is injective. Indeed, by Corollary 1.2, it suffices to check after modulo $\mathfrak{m}$. Therefore, we need to show that the perfection comparison map $(\pi_0(R)/\mathfrak{m})^\flat \simeq R^\flat \to \pi_0(R)/\mathfrak{m}$ is injective. Since everything is of Krull dimension 0, the kernel is generated by idempotents and meanwhile nilpotent, which must vanish.

Therefore, the first and second conditions make sure that the colocalization map is isomorphic on even degrees. The vanishment of odd degrees then guaranties that it is an equivalence. □

Definition 2.8. We define the following three maps in $\mathrm{CAlg}^{\wedge}_{E(k)}$:

(1) $f : E(k[t]) \to E(k[t^{\pm 1}]) \times E(k)$ as in Variant 1.6.
(2) $g : E(k)\{z^0\} \to E(k\{z^0\}^\sharp)$, which is to be defined soon
(3) $h : E(k)\{z^1\} \xrightarrow{z^1 \mapsto 0} E(k)$.

We will prove the following.

Theorem 2.9. *Let f, g, h be defined as above. Then f, g, h detect nilpotence. Moreover, $f \perp R$ if and only if R satisfies (1) in Lemma 2.7; $g \perp R$ if and only if R satisfies (2); $h \perp R$ if and only if R satisfies (3).*

Once we prove this theorem, we can use the small object argument to conclude that for any $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, we can find an extension $R \to R'$ which detects nilpotence, and such that $R' \simeq E(A)$ for some A of Krull dimension 0. We win.

The statements about f come from the previous talk. We need only to discuss the statement for g and h.

2.3. **The map g.** We begin with some notations. We use $k\{z^0\}$ to denote the symmetric polynomial. We have an equivalence $k\{z^0\} \cong \pi_0(E(k)\{z^0\})/\mathfrak{m}$. To define g, we need the following lemma.

Lemma 2.10. *Let A be a perfect k-algebra. Then the functor*

$$(\pi_0(-)/\mathfrak{m})^\sharp : \mathrm{CAlg}^{\wedge}_{E(k)} \to \mathrm{Perf}_k$$

induces an isomorphism

$$\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) \to \mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, A)$$

Proof. One can factor the map as

$$\begin{aligned}\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) &\xrightarrow{\pi_0} \mathrm{Map}_{\mathbb{T}}(\pi_0(E(k)\{z^0\}), \pi_0(E(A))) \\ &\xrightarrow{(U_{\mathbb{T}}(-)/\mathfrak{m})^\sharp} \mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, A).\end{aligned}$$

The second arrow is an isomorphism by cofreeness of Lubin-Tate theories. It suffices to show the first isomorphism. Consider the diagram

$$\begin{array}{ccccc}\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) & & \xrightarrow{\quad\simeq\quad} & & \pi_0\Omega^\infty E(A) \\ \downarrow{\scriptstyle \pi_0} & & & & \downarrow{\scriptstyle \simeq} \\ \mathrm{Map}_{\mathbb{T}}(\pi_0(E(k)\{z^0\}), \pi_0(E(A))) & \longrightarrow & \mathrm{Map}_{\mathbb{T}}(\mathbb{T}(\pi_0 E(k)), \pi_0(E(A))) & \xrightarrow{\simeq} & \pi_0 E(A)\end{array}$$

where $\mathbb{T}(\pi_0 E(k))$ is the free $\mathbb{T}$-algebra on one generator. The bottom left horizontal map is induced by precomposition with the map $\mathbb{T}(\pi_0 E(k)) \to \pi_0(E(k)\{z^0\})$, which is an isomorphism since $\pi_0 E(A)$ is $\mathfrak{m}$-adically complete and $\mathbb{T}(\pi_0 E(k))^{\wedge}_{\mathfrak{m}} \cong \pi_0(E(k)\{z^0\})$ (see [Rez09, Proposition 4.17]). The top and bottom right horizontal equivalences are because of freeness. □

Definition 2.11. We define $g : E(k)\{z^0\} \to E(k\{z^0\}^\sharp)$ as the map that associates $\mathrm{id} : E(k\{z^0\}^\sharp) \to E(k\{z^0\}^\sharp)$.

Our next aim is to show that g satisfies the desired properties.

Proposition 2.12. *We have the following properties:*

1. *The functor $- \otimes_{E(k)\{z^0\}} E(k\{z^0\}^\sharp)$ induced by g is conservative. In particular, g detects nilpotence.*
2. *For $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, we have $g \perp R$ if and only if $\pi_0(E(R^\flat)) \to \pi_0(R)$ is surjective.*

Proof. The first assertion is by reducing to $\mathrm{Mod}_{k\{z^0\}}$ and the fact that $k\{z^0\}^\sharp$ is free as a $k\{z^0\}$-module. The second assertion is because we have

$$\begin{aligned}\pi_0\mathrm{Map}(E(k\{z^0\}^\sharp), R) &\cong \pi_0\mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, R^\flat)\\ &\cong \pi_0\mathrm{Map}(E(k)\{z^0\}, E(R^\flat))\\ &\cong \pi_0(E(R^\flat))\end{aligned}$$

The first isomorphism is from the Lubin-Tate-tilt adjunction, the second is by the above lemma, and the last is by the definition of $E(k)\{z^0\}$. □

Once we check the properties of f and g, using small object argument, we obtain the following intermediate consequence.

Corollary 2.13. *Let $R \in \mathrm{CAlg}^\wedge_{E(k)}$. Then there exists a nilpotence detecting map $R \to R'$ such that*

$$E(R'^\flat) \to R'$$

induces an isomorphism on π_0 and $\pi_0(R')$ is reduced of Krull dimension 0.

2.4. **The map h.** We next consider the map $h : E(k)\{z^1\} \xrightarrow{z^1\mapsto 0} E(k)$. We need to show that h detects nilpotence. Our strategy is to convert the problem of nilpotence detection to the problem of the existence of special filtrations.

We make some preparations. Let $\alpha \in \pi_1 R$. We use $h_\alpha : R \to R//\alpha$ to denote the $\mathbb{E}_\infty$-quotient of R by α. That is, the commutative algebra map obtained by base-change along h. Recall the definition of phantom maps:

Definition 2.14. Let $\mathcal{C}$ be a rigid category. A map $f : a \to b$ is called phantom if for any $g : c \to a$ with $c \in \mathcal{C}^\omega$, $f \circ g$ is null. f is called $\otimes^m$-phantom if $f^{\otimes m}$ is phantom.

That the fiber of a map is phantom implies the map itself detects nilpotence. We will prove the following.

Proposition 2.15. *Let $R \in \mathrm{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced and $\alpha \in \pi_1(R)$. Then the fiber of h_α is $\otimes^2$-phantom. In particular, h_α detects nilpotence.*

Once we prove this, it follows that h detects nilpotence. This is because by Corollary 2.13, we can always find a map $q\colon E(k)\{z^1\} \to R$ such that R is reduced. Consider the commutative square

$$\begin{array}{ccc} E(k)\{z^1\} & \xrightarrow{q} & R \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h_\alpha} \\ E(k) & \longrightarrow & R//\alpha \end{array}$$

The top row detects nilpotence by assumption, and the right vertical map detects nilpotence by the proposition. As a consequence, h detects nilpotence as well.

We will convert the problem of $\otimes^2$-phantom to the existence of certain filtrations. We will need the following notation.

Notation 2.16. We only consider the ascending filtration. That is, the filtration indexed by $\mathbb{N}$ such that 0 is the initial object. For a symmetric monoidal stable category $\mathcal{C}$, we have the following usual functors

$$\mathcal{C} \underset{\text{ct}}{\overset{\text{colim}}{\leftarrow}} \text{Fil}(\mathcal{C}) \xrightarrow{\text{gr}} \text{Gr}(\mathcal{C}) \xrightarrow{\oplus} \mathcal{C}$$

Definition 2.17. Let $\mathcal{C}$ be a symmetric monoidal stable ∞-category. For $M \in \text{Gr}(\mathcal{C})$, we say that M is **purely even** if for all i, M_i is a finite direct sum of even suspensions of $\mathbb{1}_{\mathcal{C}}$. In this case, the **rank** of M, which we denote by $\text{rank}(M)$, is the total number of summands in $\bigoplus M_i$. For $M_\bullet \in \text{Fil}(\mathcal{C})$, we say M is purely even of rank r if $\text{gr}(M)$ is purely even of rank r. We say $M_\bullet$ is purely odd of rank r if ΣM is purely even of rank r.

Our aim is to prove the following criterion for $\otimes^2$-phantom maps.

Lemma 2.18. *Let $R \in \text{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced and let $M_\bullet \in \text{Fil}(\text{Mod}^\wedge_R)$ be purely odd. Then any map $\text{colim}M_\bullet \to R$ is $\otimes^2$-phantom.*

The lemma is based on the following result.

Proposition 2.19. *Let $R \in \text{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced. Suppose $N_1 \xrightarrow{f} N_2 \xrightarrow{g} \text{ct}(R)$ is a sequence of maps in $\text{Fil}(\text{Mod}^\wedge_R)$ such that N_1 is purely even of finite rank and N_2 is purely odd of finite rank. Then the composite $g \circ f$ is null.*

Proof. First, since N_1 is of finite rank, one can properly choose some $\Sigma^{2j}\text{ct}(R)\langle m\rangle$ such that we have a cofiber sequence

$$\Sigma^{2j}\text{ct}(R)\langle m\rangle \to N_1 \to N_1'$$

where $\text{rank}(N_1') = \text{rank}(N_1) - 1$. By induction along the cofiber sequence, we reduce to the case where $\text{rank}(N_1) = 1$.

When $\text{rank}N_1 = 1$, $g \circ f$ determines a point in $\text{Map}(N_1, \text{ct}(R)) \simeq \text{Map}(\text{colim}N_1, R)$. Since N_1 is of rank 1, this lies in some even homotopy group of R. Note that when R is reduces, $R \to R \otimes \mathbb{Q}$ induces injection on homotopy groups, it therefore suffices to show in the rational case.

We consider the following diagram

$$\begin{array}{ccccc} N_1^{\otimes m} & \xrightarrow{f^{\otimes m}} & N_2^{\otimes m} & \xrightarrow{g^{\otimes m}} & \text{ct}(R)^{\otimes m} \\ \downarrow & & \downarrow & & \downarrow \\ {N_1}_{h\Sigma_m}^{\otimes m} & \longrightarrow & {N_2}_{h\Sigma_m}^{\otimes m} & \longrightarrow & \text{ct}(R)_{h\Sigma_m}^{\otimes m} \end{array}$$

where m is the rank of N_2. In the rational case, one can compute that the outside vertical maps are equivalence and the middle map is zero. Therefore, $(g \circ f)^{\otimes m}$ is null. Now since the even homotopy group of R is reduced, $g \circ f$ is null. □

Proof of Lemma 2.18. The filtered colimit of $\otimes^m$-phantom maps is still $\otimes^m$-phantom. Therefore, we reduce to the case where $M_\bullet$ is of finite rank. Any map $\text{colim}M_\bullet \to R$ is the colimit of $f : M_\bullet \to \text{ct}(R)$, where f can be decomposed as

$$M_\bullet^{\otimes 2} \xrightarrow{f\otimes \text{id}} M_\bullet \xrightarrow{f} \text{ct}(R)$$

which is null by the above lemma and in particular phantom. □

Let $\mathrm{Fib}(h_\alpha) \to R \xrightarrow{h_\alpha} R//\alpha$ be the fiber sequence. To prove Proposition 2.15, it suffices to construct a filtration on this fiber sequence such that it is purely odd on $\mathrm{Fib}(h_\alpha)$. We can first construct a filtration on $R//\alpha$.

Proposition 2.20. *Given a $T(n)$-local commutative algebra R, let $\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}$ denote the free filtered commutative R-algebra with an element $z^{1\langle 1\rangle}$ in double degree $(1,1)$. Then, for any $\alpha \in \pi_1 R$, the image of α in filtration degree 1 of $\mathrm{ct}(R)$ determines a map of filtered objects. We define $R^\alpha_\bullet$ as the pushout*

$$\begin{array}{ccc} \mathrm{ct}(R)\{z^{1\langle 1\rangle}\} & \xrightarrow{\alpha} & \mathrm{ct}(R) \\ {\scriptstyle 0}\downarrow & \ulcorner & \vdots \\ \mathrm{ct}(R) & \dashrightarrow & R^\alpha_\bullet \end{array}$$

Then the underlying object of $R^\alpha_\bullet$ is $R//\alpha$ and the fiber of $\mathrm{ct}(R) \to R^\alpha_\bullet$ is purely odd.

Proof. The first statement follows immediately because we can apply colim to the defining diagram. For the purely odd part, note that the graded pieces of $\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}$ is computed by

$$\mathrm{gr}(\mathrm{ct}(R)\{z^{1\langle 1\rangle}\})_i \cong (\Sigma R)^{\otimes i}_{h\Sigma_i}$$

Applying the graded pieces functor to the defining diagram, we obtain

$$\begin{array}{ccc} \mathrm{gr}(\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}) & \xrightarrow{0} & \mathrm{gr}(\mathrm{ct}(R)) \\ {\scriptstyle 0}\downarrow & \ulcorner & \downarrow \\ \mathrm{gr}(\mathrm{ct}(R)) & \longrightarrow & \mathrm{gr}(R^\alpha_\bullet) \end{array}$$

By which we compute

$$\mathrm{gr}(R^\alpha_\bullet)_i \cong (\Sigma^2 R)^{\otimes i}_{h\Sigma_i}$$

The purely odd part then follows by checking the rank of graded pieces. □

3. Proof of Theorem A

We now prove the Theorem A of [BSY22]. We recall the definition of Nullstellensatzian object in a general category.

Definition 3.1. We say that a presentable ∞-category $\mathcal{C}$ is **Nullstellensatzian** if every compact and non-terminal object admits a map to the initial object. We say that an object R in a presentable ∞-category $\mathcal{C}$ is Nullstellensatzian if R is non-terminal and $\mathcal{C}_{R/}$ is Nullstellensatzian.

The classical Hilbert Nullstellensatz states that the Nullstellensatzian objects in CRing are exactly algebraic closed fields. We will prove the following:

Theorem 3.2. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be a nonzero object. R is Nullstellensatzian if and only if $R = E(L)$ for some algebraic closed field L.*

Bofore going on, we explain a property in which the Nullstellensatzian object in a presentable category mimicing algebraic closed fields.

Proposition 3.3. *Let $\mathcal{C}$ be a presentably symmetric monoidal ∞-category and let $R \in \mathrm{CAlg}(\mathcal{C})$ be Nullstellensatzian. Suppose W_1, W_2 are compact R-modules and $T \in \mathrm{CAlg}_R(\mathcal{C})$, we let $0 : R\{W\} \to T$ denote the R-algebra map induced the zero map $W \to T$. Then, for any algebra map $P\colon R\{W_1\} \to R\{W_2\}$, the following are equivalent:*

(1) There exists a nonzero $T \in \mathrm{CAlg}_R(\mathcal{C})$ and a factorization in $\mathrm{CAlg}_R(\mathcal{C})$:

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ & \searrow^{0} & \downarrow \\ & & T \end{array}$$

(2) There exists a factorization in $\mathrm{CAlg}_R(\mathcal{C})$:

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ & \searrow^{0} & \downarrow \\ & & R \end{array}$$

Proof. It suffices to show that (1) implies (2). Since W_1, W_2 are compact, it follows that the algebras $R\{W_1\}$ and $R\{W_2\}$ are also compact in $\mathrm{CAlg}_R(\mathcal{C})$. By assumption, we have a factorization

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ {\scriptstyle 0}\downarrow & & \downarrow \\ R & \longrightarrow & T \end{array}$$

Denote

$$Q := R \otimes_{R\{W_1\}} R\{W_2\}$$

Then Q is compact in $\mathrm{CAlg}_R(\mathcal{C})$ and $Q \neq 0$. Thus, the condition that R is Nullstellensatzian inmplies there exists a retract $R \to Q \to R$ which gives the desired factorization. □

One can think of P as an equation. The above proposition amounts to say that if R is Nullstellensatzian, then an equation has a solution in R if and only if it has a solution in some of its extension, which is exactly a defining property of an algebraic closed field.

Corollary 3.4. *Fix the notations as above, then the induced map*

$$\mathrm{map}_{\mathcal{C}}(W, R) \to \mathrm{map}_{\mathcal{C}}(W, T)$$

on mapping spectra is injective on homotopy groups.

Proof. Suppose $x \in \ker(\pi_k \mathrm{map}_{\mathcal{C}}(W, R) \to \pi_k \mathrm{map}_{\mathcal{C}}(W, T))$. x defines a map of algebra

$$x : R\{\Sigma^k W\} \to R$$

such that the composite

$$R\{\Sigma^k W\} \xrightarrow{x} R \to T$$

is homotopic to 0. By the proposition, $x : R\{\Sigma^k W\} \to R$ is homotopic to $0 : R\{\Sigma^k W\} \to R$. □

Remark 3.5. Suppose we additionally require $\mathcal{C}$ to be rigid, then the above corollary has a dual form, that means, the map

$$W \otimes R \to W \otimes E(L)$$

is injective on homotopy groups.

Now we can prove Theorem 3.2. Since we have a description of the essential image of the Lubin-Tate functor, it suffices to check that any Nullstellensatzian object in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ satisfies these properties. To begin with, we have:

Lemma 3.6. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be Nullstellensatzian. Then R is even.*

Proof. By Corollary 2.2, there exists a map $R \to E(L)$ for some algebraic closed field L. Choose V_n to be the generalized Smith-Toda complex. By Corollary 3.4 and Remark 3.5, the map

$$V_n \otimes R \to V_n \otimes E(L)$$

is injective on homotopy groups. Therefore $V_n \otimes R$ is even, which implies R is even. □

As a consequence, on can choose $v_0, \cdots, v_{n-1} \in \pi_* R$ and define

$$K := R/v_0 \otimes_R \cdots \otimes_R R/v_{n-1}$$

Lemma 3.7. *K is a compact R module.*

Proof. K is generated under finite colimit of R, so it is dualizable, which implied $K \otimes V_n$ is compact. Since K is a retract of $K \otimes V_n$, K is compact. □

Now it remains to show the following.

Proposition 3.8. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be Nullstellensatzian. Then the graded ring $\pi_* K$ is even periodic and $\pi_0(K)$ is an algebraically closed field.*

Proof. We first show that $\pi_* K$ is even and commutative. Choose a map $R \to E(L)$ for some algebraically closed field L. Since K is compact, the extension

$$\iota : K \to K \otimes_R E(L) \simeq E(L)//\mathfrak{m}$$

is injective on homotopy groups. This implies that $\pi_* K$ is even commutative.

To show the remaining parts of the proposition, we show that any even graded polynomial $f \in \pi_*(K)[t_1, \cdots, t_l]$ has a solution in $\pi_* K$. One can regard f as a natural transformation

$$f : \prod_{i=1}^{l} \pi_{2d_i}(- \otimes_R K) \to \pi_{2d}(- \otimes_R K)$$

from $\mathrm{CAlg}_R^{\wedge}$ to $\mathrm{Fun}(\mathbb{Z}, \mathrm{Set})$, where we fix $\Sigma d_i = d$. By Yoneda, f is represented by

$$P_f : R\{\Sigma^{2d} K^{\vee}\} \to R\{\bigoplus_{i=1}^{l} \Sigma^{2d_i} K^{\vee}\}$$

The condition that f has a solution is equivalent to say that we have a factorization

$$\begin{array}{ccc} R\{\Sigma^{2d}K^\vee\} & \xrightarrow{P_f} & R\{\bigoplus_{i=1}^{l}\Sigma^{2d_i}K^\vee\} \\ & \searrow^{0} & \downarrow \\ & & \pi_*(K) \end{array}$$

Note that through the extension $K \to K \otimes_R E(L) \simeq E(L)//\mathfrak{m}$, f always has a solution in $\pi_*(E(L)//\mathfrak{m}) \cong L[u^{\pm 1}]$ since L is algebraic closed. It follows from Proposition 3.3 that f has a solution in $\pi_*(K)$.

Therefore, if we take $f = at - 1$, that f has a solution means $\pi_0(K)$ is a field. Taking $f = t_1t_2 - 1$ with t_1 in degree 2 shows that π_*K is periodic, and taking f to be any ungraded polynomial shows that π_0K is algebraic closed. □

Combining the above consequences and the description of the essential image of $E(-)$, we proved Theorem 3.2.

Remark 3.9. One can define Nullstellensatzian for general regular cardinalities. The similar statement of Theorem 3.2 holds true for any regular cardinality.

4. Constructable spectra

We briefly introduce the constructable spectra theory.

The Theorem D shows that there are enough geometric points in chromatic world, and the Theorem A implies that all the geometric points are closed. It brings a natural way for us to study the geometry of these chromatic rings. The first step is to attempt to define a topological space $\mathrm{Spec}(R)$ for $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$.

We introduce the following definition.

Definition 4.1. Let $\mathcal{C}$ be the category of products of algebraically closed fields. These fields are equipped with a formal group of height n, which is unique up to isomorphism. We define the **constructible spectrum** functor

$$\mathrm{Spec}_{T(n)}(-) : \mathrm{CAlg}(\mathrm{Sp}_{T(n)})^{\mathrm{op}} \to \mathrm{CHaus}$$

as the left Kan extension:

$$\begin{array}{ccc} \mathcal{C}^{\mathrm{op}} & \xrightarrow{\mathrm{Spec}(-)} & \mathrm{CHaus} \\ & \searrow^{E(-;-)} \quad \nearrow_{\mathrm{Spec}_{T(n)}(-)} & \\ & \mathrm{CAlg}(\mathrm{Sp}_{T(n)})^{\mathrm{op}} & \end{array}$$

Theorem 4.2. *The constructible spectrum functor* $\mathrm{Spec}_{T(n)}(-)$ *enjoys the following properties:*

1. $\mathrm{Spec}_{T(n)}(R)$ *is empty if and only if* $R = 0$.
2. *If* L *is an algebraically closed field, then* $\mathrm{Spec}_{T(n)}(E(L))$ *is a point.*
3. *For every point* $q \in \mathrm{Spec}_{T(n)}(R)$*, there exists an algebraically closed field* L *and a map* $R \to E(L)$ *such that the image of the map*

$$\{*\} \cong \mathrm{Spec}_{T(n)}(E(L)) \to \mathrm{Spec}_{T(n)}(R)$$

is $\{q\}$.

(4) *A subset* $U \subset \mathrm{Spec}_{T(n)}(R)$ *is closed if and only if there exists a map* $R \to S$ *such that* U *is the image of the map*

$$\mathrm{Spec}_{T(n)}(S) \to \mathrm{Spec}_{T(n)}(R)$$

(5) *Given a span* $S \leftarrow R \to T$*, the natural comparison map*

$$\mathrm{Spec}_{T(n)}\left(S \otimes_R T\right) \to \mathrm{Spec}_{T(n)}(S) \times_{\mathrm{Spec}_{T(n)}(R)} \mathrm{Spec}_{T(n)}(T)$$

is surjective.

These reflect basic properties of the classical Zariski spectrum.

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LECTURE 6

TYPED BY AOWEN FAN

In this lecture, we cover the contents of [BSY22, Section 8]. We discuss the application of the Nullstellensatz. We will use the Nullstellensatz theorem to explain orientations of Lubin-Tate theory and compute the strict picard spectrum of algebraic closed Lubin-Tate theories.

1. Pullback squares

Recall that by the first theorem in the last talk, for any none-zero $R \in \mathrm{CAlg}(\mathrm{Sp}T(n))$, there exists a map $R \to E(L)$ in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. We can base change their modules in $\mathrm{Sp}_{T(n)}$ along this extension. The main theorem of this talk is

Theorem 1.1. *Suppose L is an algebraic closed field and $R \in \mathrm{CAlg}^{\wedge}_{E(L)}$. Then the base change $\mathrm{picMod}^{\wedge}_{E(L)} \to \mathrm{picMod}^{\wedge}_{R}$ admits a retract.*

We will heavily use the group of unit functor $\mathfrak{gl}_1 \colon \mathrm{CAlg} \to \mathrm{Sp}_{\geqslant 0}$. Recall that $\Omega\mathrm{pic}(R) = \mathfrak{gl}_1(R)$.

Moreover, we have $\mathfrak{gl}_1 E(-) : \mathrm{Perf}_k \to \mathrm{Sp}_{\geqslant 0}$ with $\mathfrak{gl}_1\pi_0 E(-)/m$ being exactly the group of units. We define $\mathfrak{sl}_1^E \in \mathrm{Sp}_{\geqslant 0}$ as the fiber of $\pi_0(-)/\mathfrak{m} : \mathfrak{gl}_1 E(-) \to \mathfrak{gl}_1(\pi_0 E(-)/\mathfrak{m})$

Proposition 1.2. *Suppose that we have a pullback in Perf_k.*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

such that $B \oplus C \to D$ is a surjection. Then after applying $\mathfrak{sl}_1^E$, we obtain a pullback square in $\mathrm{Sp}_{\geqslant 0}$:

$$\begin{array}{ccc} \mathfrak{sl}_1^E A & \longrightarrow & \mathfrak{sl}_1^E B \\ \downarrow & & \downarrow \\ \mathfrak{sl}_1^E C & \longrightarrow & \mathfrak{sl}_1^E D \end{array}$$

Proof. The construction of $\mathfrak{sl}_1^E$ includes applying the Lubin-Tate functor, the group of unit functor and the fiber to a given pullback square. All of these operators preserve pullbacks. To eliminate the "$\geqslant 0$" part, we need to show that $\pi_0(\mathfrak{sl}_1^E C) \oplus \pi_0(\mathfrak{sl}_1^E B) \to \pi_0(\mathfrak{sl}_1^E D)$ is a surjection. This follows from the surjectivity of $B \oplus C \to D$ and the complete filtration powered by $\mathfrak{m}$ on both sides. □

Corollary 1.3. *Suppose that we have a pullback square*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

such that

(1) $B \oplus C \to D$ is a surjection.
(2) $B^\times \oplus C^\times / A^\times \to D^\times$ admits a retract.
(3) $A \to B$ and $A \to C$ admit retracts.

Then $\mathfrak{gl}_1(A) \to \mathfrak{gl}_1(D)$ admits a retract.

Proof. Note that (3) implies that we have a retract for

$$\mathfrak{gl}_1 E(A) \to \mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C)$$

Therefore, it suffices to find a retract for

$$\mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C) \to \mathfrak{gl}_1 E(D)$$

This follows from the above proposition, which implies that we have a pullback square

$$\begin{array}{ccc} \mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C) & \longrightarrow & B^\times \coprod_{A^\times} C^\times \\ \downarrow & & \downarrow \\ \mathfrak{gl}_1 E(D) & \longrightarrow & D^\times \end{array}$$

and the fact that (2) implies right vertical arrow admits a splitting. □

2. Hahn series

Definition 2.1. Let Γ be a totally ordered commutative monoid and A be an integral domain. We define $A[\![t^\Gamma]\!]$ to be the set of functions $\Gamma \to A$ that have well-ordered support. It admits an obvious addition and a multiplication with formula

$$f \cdot g(x) = \sum_{y+z=x} f(y)g(z)$$

under which $A[\![t^\Gamma]\!]$ becomes a ring.

Lemma 2.2. *(1) $f \in A[\![t^\Gamma]\!]$ is invertible if and only if $\nu(f)$ is invertible and $\operatorname{arc}(f)$ is invertible, where the two maps are given by*

$$\begin{aligned} \nu \colon A[\![t^\Gamma]\!]^* &\to \Gamma & \qquad \operatorname{arc} \colon A[\![t^\Gamma]\!]^* &\to A^* \\ f &\mapsto \min_{f(\gamma)\neq 0} \gamma & f &\mapsto f(\nu(f)) \end{aligned}$$

(2) If A is a field, Γ is a group, then $A[\![t^\Gamma]\!]$ is a field.
(3) Let L be an algebraic closed field and Γ be a $\mathbb{Q}$-module, then $L[\![t^\Gamma]\!]$ is an algebraic closed field.

Corollary 2.3. *Let L be an algebraic closed field and F be a non-zero L-algebra. Suppose that Γ is a totally ordered $\mathbb{Q}$-module with $|\Gamma| > |F|$, then there exists an L-algebra map $F \to L[\![t^\Gamma]\!]$.*

Proof. By modulo the maximal ideal and taking the algebraic closure, we reduce to the case where F is algebraic closed. We have

$$|L[\![t^\Gamma]\!]| \geqslant |\Gamma| > |F|$$

Therefore, we conclude $\mathrm{tr}_L(|L[\![t^\Gamma]\!]|) > \mathrm{tr}_L(|F|)$ and there exists an inclusion $F \to L[\![t^\Gamma]\!]$. □

Lemma 2.4. *Let $A \in \mathrm{Perf}_k$ be a non-zero domain and Γ be a totally ordered monoid such that $p : \Gamma \to \Gamma$ is an isomorphism. Then the map*

$$\mathfrak{gl}_1 E(A) \to \mathfrak{gl}_1 E(A[\![t^\Gamma]\!])$$

admits a retract.

Proof. This follows from applying Corollary 1.3 to the pullback square

$$\begin{array}{ccc} A & \longrightarrow & A[\![t^{\Gamma_{\leqslant 0}}]\!] \\ \downarrow & & \downarrow \\ A[\![t^{\Gamma_{\geqslant 0}}]\!] & \longrightarrow & A[\![t^\Gamma]\!] \end{array}$$

and the description of the invertible elements in $A[\![t^\Gamma]\!]$. □

Now we can prove our main theorem.

Theorem 2.5. *Suppose L is an algebraic closed field and $R \in \mathrm{CAlg}^\wedge_{E(L)}$. Then the base change $\mathrm{picMod}^\wedge_{E(L)} \to \mathrm{picMod}^\wedge_R$ admits a retract.*

Proof. By the existence of extension $F \to L[\![t^\Gamma]\!]$ and the Nullstellensatz theorem, we reduce to the case where $R = L[\![t^\Gamma]\!]$. We consider the following square

$$\begin{array}{ccc} \Sigma\mathfrak{gl}_1 E(L) & \longrightarrow & \Sigma\mathfrak{gl}_1 E(L[\![t^\Gamma]\!]) \\ \downarrow & & \downarrow \\ \mathrm{pic}E(L) & \longrightarrow & \mathrm{pic}E(L[\![t^\Gamma]\!]) \end{array}$$

This square is a pushout square since they admit isomorphic cofibers. By Lemma 2.4, the above arrow has a retract, which yields a retract for the bottom arrow. □

3. Orientations and Thom spectra

Definition 3.1. Let $\mathcal{C}$ be a symmetric monoidal category and $X \to \mathrm{pic}(\mathcal{C})$ be a map in $\mathrm{Sp}_{\geqslant 0}$. We define the *Thom spectrum* of f to be $Mf := \mathrm{colim}_X f \in \mathrm{CAlg}(\mathcal{C})$, where the colimit is taken in $\mathcal{C}$.

According to [And+14], Mf satisfies a specific universal property. That is, the space of orientations from Mf to R can be identified with the lifting of f to trivial R-modules.

Proposition 3.2. *Let $\mathcal{C}$ be a presentable symmetric monoidal category. Then the followings are equivalent.*

(1) $f : X \to \mathrm{pic}\mathcal{C}$ is null.

(2) There is a map $Mf \to \mathbb{1}_{\mathcal{C}}$ *in* $\mathrm{CAlg}(\mathcal{C})$.
(3) $\mathrm{pic}(\mathcal{C}) \to \mathrm{pic}(\mathrm{Mod}_{Mf}(\mathcal{C}))$ *admits a retract.*

Corollary 3.3. *Let* L *be an algebraic closed field and* $f\colon X \to \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})$ *be a map in* $\mathrm{Sp}_{\geqslant 0}$. *The the followings are equivalent:*

(1) f *is null.*
(2) There is a map $Mf \to E(k)$.
(3) Mf *is nonzero.*
(4) $Mf \simeq E(L)[X]$.

4. Strict Picard group of Lubin-Tate theory

The above discussion can be used to compute the strict Picard group of Lubin-Tate theory.

Definition 4.1. Let $\mathcal{C}$ be a presentable symmetric monoidal category. The strict Picard spectrum of $\mathcal{C}$ is defined as

$$\mathbb{G}\mathrm{pic}(\mathcal{C}) = \underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathcal{C})) \simeq \tau_{\geqslant 0}\mathrm{map}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathcal{C}))$$

Proposition 4.2. *Let* L *be an algebraically closed field and let* H *be a* p*-torsion abelian group. There is a natural equivalence of connective spectra*

$$\underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geq 0}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})) \simeq \Sigma^{n+1} H^*$$

where $H^* = \mathrm{Hom}(H, \mathbb{Q}/\mathbb{Z})$ *stands for Pontryagin dual of* H.

Proof. (Sketch of proof) We first show that $\pi_m \mathrm{Map}_{\mathrm{Sp}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})) \cong 0$ for $m \neq n+1$. It suffices to show that a class

$$f \in \pi_0(\mathrm{Map}_{\mathrm{Sp}}(\Sigma^m H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)}))$$

is null with $0 \leqslant m \leqslant n$. By Corollary 3.3, it suffices to show that $Mf \neq 0$, which is by argument in [HL13]. Now we need to identify the functor $\pi_n \underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geq 0}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)}))$ with the Pontryagin dual $(-)^*$. This can be check on the set level, which is given by

$$\begin{aligned}[\Sigma^{n+1} H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})] &\cong [\Sigma^n H, \mathfrak{gl}_1(E(L))] \cong [E(L)[B^n H], E(L)]_{\mathrm{CAlg}^{\wedge}_{E(L)}} \\ &\cong [E(L)^{H^*}, E(L)]_{\mathrm{CAlg}^{\wedge}_{E(L)}} \cong H^*\end{aligned}$$

□

Theorem 4.3. *Let* L *be an algebraically closed field of characteristic* p *and assume that the height* $n \geqslant 1$ *Then we have an equivalence of connective* $\mathbb{Z}$*-modules*

$$\mathbb{G}\mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(k)}) \simeq \Sigma^{n+2}\mathbb{Z}_p \oplus \Sigma L^{\times}$$

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